

WORK SHEET - 04

WORK POWER ENERGY

HINTS & SOLUTIONS

Q.1 (B)

$$W_{\text{motor}} + W_{\text{fr}} = \Delta K = \frac{1}{2} mv^2$$

as seen from belt's frame,

$$W_{\text{friction}} = \text{heat dissipated} = -\frac{1}{2} mv^2$$

$$W_{\text{motor}} = mv^2$$

Q.19 (B)

$$W_T + W_g = \Delta KE$$

$$40 \times (0.5 - 0.1) - 0.8 \times 10 \times 0.4 = \frac{1}{2} mv^2 - 0$$

Q.20 (A)

Contact between them will be maintained until the spring reaches its natural length.

Q.21 (BCD)

$$\text{Work done by } F_2 \text{ is } W_2 = 30 \times 6 = 180 \text{ J}$$

$$\text{Work done by } F_3 \text{ is } W_3 = 15 \times \left(\frac{2\pi \times 6}{4} \right) = 45\pi \text{ J}$$

$\vec{F}_1$ is a central force with a constant magnitude. So its work done depends on initial & final positions only.]

Q.24 (D)

Force on the particle will be given as $\vec{F} = \nabla U$

$$= \frac{\partial U}{\partial x} \hat{i} - \frac{\partial U}{\partial y} \hat{j} = -(6xy^2 + 6) \hat{i} - (6x^2y) \hat{j}$$

Now, for acceleration at $(t=0)$ and $x=1, y=1$

$$\vec{a} = \frac{\vec{F}}{M} = 12\hat{i} - 6\hat{j}$$

$$|\vec{a}| = \sqrt{180} \text{ m/s}^2$$

Particle is at rest at $x=1, y=1$, then

$$P.E. + K.E. = M.E.$$

$$U(1, 1) + K.E.(1, 1) = M.E.$$

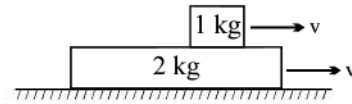
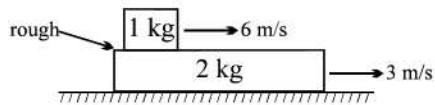
$$\Rightarrow M.E. = 9 \text{ J}$$

$$\Delta U + \Delta K = \Delta W$$

$$\Delta W = \Delta U = U(0, 0) - U(1, 1)$$

$$\Delta W = -9 \text{ J}$$

Q.25 (A) Q; (B) P,S; (C) P,T; (D) R, Q



When relative motion stop, both block will move with same velocity v , momentum conservation gives,

$$1 \times 6 + 2 \times 3 = 3v$$

$$\Rightarrow v = 4 \text{ m/s}$$

Work done by friction on block = change in K.E.

$$(A) \Rightarrow W_{f1} = \frac{1}{2} \times 1 \times (4)^2 - \frac{1}{2} \times 1 \times (6)^2 = -10 \text{ J}$$

$$(B) \quad W_{f2} = \frac{1}{2} \times 2 \times (4)^2 - \frac{1}{2} \times 2 \times (3)^2 = 7 \text{ J}$$

$$(C) \quad 4 \times 2 - 3 \times 2 = 2 \text{ N-s}$$

$$(D) \quad \text{Now total work done by friction} = (-10 + 7) = -3 \text{ J}$$

$$\therefore \text{Total change in K.E.} = -3 \text{ J}$$

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